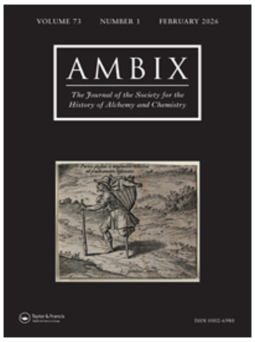

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Introduction: Computational Approaches to the Histories of Alchemy and Chemistry

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The promise of computational methods opens new epistemic horizons. Yet, as we discuss in this introduction to the special issue on computational approaches to the histories of alchemy and chemistry, it also brings new epistemic responsibilities. We situate this work within a broader digital and computational history, exploring its relationship with earlier traditions of quantitative history as well as with developments in digital humanities and computational humanities. We distinguish between digital, computational, algorithmic, and AI-driven approaches and we clarify methodological and epistemological concerns that keep the historian firmly in the loop. Claims about computational methods often imply that they enable entirely new modes of inquiry. Yet such sweeping claims can also obscure the limitations and conditions under which these methods operate. This introduction therefore explores both the opportunities and the constraints of computational approaches that more celebratory accounts sometimes overlook. It introduces the individual contributions to the special issue and concludes with a call for responsibility in the adoption of computational methods. We argue for making the histories of alchemy and chemistry more critical, global, and reflexive, while remaining attentive to the assumptions, limitations, and implications of the computational tools we employ.

New epistemic horizons, new epistemic responsibilities

The histories of alchemy and chemistry stand at a pivotal juncture. In a field traditionally defined by the close reading of manuscripts, the meticulous reconstruction

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of laboratory practices, and the narrative synthesis of individual lives and ideas, computational methods open a new epistemic horizon.[1] This special issue of *Ambix* emerges from a growing recognition that those methods are more than auxiliary tools to historical scholarship – they are transformative, enabling historians to ask new questions, uncover hidden patterns, and reframe long-standing debates in ways previously imagined but impracticable.

In a 2022 special issue of *Ambix* on the "Chemical Humanities," historians Megan Piorko, Marieke Hendriksen and Simon Werrett suggested that chemical metaphors such as catalysis might be useful for understanding trends in the history of alchemy:

> consider the use of digital tools to study Decknamen … [T]he great time and effort needed to do this on a large scale might be obviated by using digital tools, capable of scanning large texts in an instant to reveal, potentially, the clusters of meanings connected to different terms. Hence a process that could occur "naturally" is accelerated through the inclusion of a catalyst. To push the analogy further: what other historical research processes might be catalysed using digital or analogue techniques?[2]

While historical research processes could certainly be catalysed by using computational techniques in tandem with analogue ones, we contend that the catalysis analogy is only partially descriptive of what such techniques can do. Articles in the present special issue show that beyond accelerating "naturally occurring" research processes, computational methods can also open new pathways of inquiry for the field.

Isis, the journal of the History of Science Society, devoted a "Focus" section to computational history and philosophy of science in 2019 – fatefully just ahead of OpenAI's wide release of ChatGPT 3 to the general public in June 2020.[3] Since then, artificial

1 Guillermo Restrepo, "Computational History of Chemistry," *Bulletin for the History of Chemistry* 47 (2022): 91–106; Sarah A. Lang, "(Doing) Computational History: The Role of Data Work in Computational Approaches," *Histories* 6, no. 2 (2026): 26, https://doi.org/10.3390/histories6020026.

2 Megan Piorko, Marieke Hendriksen and Simon Werrett, "Alchemical Practice: Looking Towards the Chemical Humanities," *Ambix* 69, no. 1 (2022): 14.

3 Manfred Laubichler, Jane Maienschein, and Abraham Gibson, "Introduction," *Isis* 110, no. 3 (2019): 497–501. OpenAI released GPT-3, the language model underlying later chatbot services, in 2020. ChatGPT, the well-known chatbot fine-tuned from a GPT-3.5 model, was released only in late 2022, attracting considerable public attention and quickly becoming almost synonymous with large language models in everyday usage. It is important to note, however, that the technological breakthrough associated with ChatGPT was not the creation of a fundamentally new type of model. Rather, it consisted in making a language model publicly accessible and sufficiently useful for a broad audience. The underlying technology had already existed and been under development for some time, but had not yet reached a stage at which it attracted significant interest beyond specialist researchers. For more information on the public hype surrounding recent AI technologies, see Arvind Narayanan and Sayash Kapoor, *AI Snake Oil: What Artificial Intelligence Can Do, What It Can't, and How to Tell the Difference* (Princeton, NJ: Princeton University Press, 2024); Emily M. Bender and Alex Hanna, *The AI Con: How to Fight Big Tech's Hype and Create the Future We Want* (New York: Harper, 2025). On the impacts and socio-technical situatedness of AI see Kate Crawford, *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* (New Haven, CT: Yale University Press, 2021). The broad adoption of AI across society and academia has also been met by criticism, for example Olivia Guest et al., "Against the Uncritical Adoption of 'AI' Technologies in Academia," Zenodo (5 September 2025), https://doi.org/10.5281/zenodo.17065099; Sarah Lang, "Critical Concerns for Using LLMs in the (Computational) Humanities and Beyond," in *Large Language Models for the History, Philosophy, and Sociology of Science: Reflections from a Field in Motion*, ed. Arno Simons et al. (Bielefeld: transcript, 2026), 35–50, https://www.transcript-verlag.de/978-3-8376-7994-6/understanding-science-with-large-language-models/?c=313000027 (accessed 3 May 2026).

intelligence (AI) chatbots have shaken the foundations of practically every sector of culture, including religion.[4] The best practices cited in the *Isis* Focus section remain generally consistent with those observed by researchers featured in the present special issue.[5] Specifically, two of the *Isis* articles referred to the workflow for computational studies in the history of science. An article by computational historians of science, Julia Damerow and Dirk Wintergrün, offers a detailed methodological discussion of the "Research data life cycle" generally applicable to computational history.[6] This process has six steps: 1. exploration of data sources; 2. acquisition of datasets; 3. dataset consolidation, cleaning, enrichment; 4. analysis of data corpus; 5. extension/adjustment of datasets; 6. Publication. It is noteworthy that steps 2 through 5 are iteratively looped in their diagram. Although they present a thorough examination of methodologies for computational history research, Damerow and Wintergrün hasten to add that "no workflow for gathering, storing, and analysing data fits all projects; to some extent, for every project the researchers will have to develop their own methods and processes."[7] This observation aligns with our experience, as will be demonstrated in the following two sections. Historians Abraham Gibson and Cindy Ermus also discuss a similar six-step workflow for computational history studies but point out that "[d]espite dreams of machine-like exactitude, every step in the process drips with value-laden judgment calls."[8] In this article, we address key concerns arising from the application of computational techniques to the history of science generally, and to the histories of alchemy and chemistry in particular.

We introduce terminological clarifications to distinguish among these techniques and to set the stage for their applications in specific studies. We conclude by outlining our perspective on computational history in light of the contributions to this special issue, as well as its broader implications for the computational humanities.

Concerns about computational methods

In our experience, some historians remain wary of what they perceive as the "decontextualisation" of historical sources, the reduction of rich, nuanced texts

4 For instance, Pope Leo's historic 2026 encyclical declares that the unregulated use of artificial intelligence in commercial and military contexts poses the grave threat of moral and spiritual harm, alters the conditions for exercising justice and humanity at a foundational level, and thereby renders the Catholic Church's centuries-old "just war" theory officially and irreversibly obsolete.

5 S. Lang, "(Doing) Computational History," 26.

6 Julia Damerow and Dirk Wintergrün, "The Hitchhiker's Guide to Data in the History of Science," *Isis* 110, no. 3 (2019): 514. On page 514, they note that their diagram: "plugs into the research life cycle outlined by Leganovic et alia … However, other parts of the research life cycle — especially the 'Annotieren' ("annotate") step — might themselves produce data and can therefore be part of the research data life cycle as well." They reaffirm the iterative nature of "human in the loop" methods of computational history.

7 Damerow and Wintergrün, "The Hitchhiker's Guide," 514.

8 Abraham Gibson and Cindy Ermus, "The History of Science and the Science of History: Computational Methods, Algorithms, and the Future of the Field," *Isis* 110, no. 3 (2019): 558–59. Those six steps are: 1. planning, 2. collecting, 3. preprocessing, 4. processing, 5. visualization, and 6. reflection. The authors' discussion of the steps is tightly bound to the specifics of their project, thus valuable as a first-hand view of concrete decision-making processes. The description of the steps themselves, however, is quite general and does not present any articulated standardised methodology.

to statistical patterns and the risk of algorithmic bias.[9] These concerns are legitimate, and one of the aims of this special issue is to confront these challenges head-on, demonstrating how computational approaches can be methodologically reflexive, historically sensitive, and epistemologically self-aware. In doing so, we seek to show that computational history is not a departure from the humanities, but a continuation – indeed, an intensification – of its established traditions, newly equipped with a set of rapidly evolving tools for discovery. It is important, therefore, to acknowledge the broader cultural context within which our discussion is situated: the explosive growth of AI in arenas ranging from home appliances to cyberwarfare and every sector of the knowledge economy. No measure appears sufficient to address the scale and intensity of this development. At best we will broaden our sources of historical information and render them accessible to analysis and new understandings.

Another aim of this special issue of *Ambix* is to demonstrate what computational methods can contribute to the histories of alchemy and chemistry – not as a replacement for traditional scholarship, but as a powerful complement to its methodological tools. We also seek to connect the field to wider discussions surrounding the computational history of science and computational humanities more broadly.[10] The articles collected here are some of the fruits of that conversation, developed independently but united by a common methodological sensibility: the belief that computation, when grounded in rigorous historical inquiry, can help us access a deeper understanding of the past, rather than a shallower one, as those wary of computational approaches often fear. These methods allow us to move beyond history at the microlevel, focusing on single texts or individuals to explore large-scale transformations in knowledge, practise, and institutions. They enable us to model the diffusion of ideas, trace the evolution of terminology, map the social networks of scientific communities, and reveal the geopolitical contours of scientific discovery. Yet it would be wrong to assume that computational history is a mere data-driven exercise, detached from the historical sources we study. On the contrary, it demands an intense, often laborious engagement with primary sources and extends the work of traditional scholarship. Selection, analysis, curation, and annotation inform the processes of digitisation and construction of training sets for advanced computational techniques– without these interwoven processes, reliable outcomes of computational analysis could not be produced. The results of

[9] Such views have been voiced for instance at conferences sponsored by the Society for the History of Alchemy and Chemistry (Oxford 2022; Philadelphia 2025), the International Consortium for the History of Chemistry (Vilnius 2023; Valencia 2025), and the International Society for the Philosophy of Chemistry (Lille, 2022; Buenos Aires, 2023).

[10] With computational methods, we refer to methods from the Computational Humanities (CH), a specialised subset of Digital Humanities (DH) that employs algorithmic and quantitative approaches such as deep learning for large-scale macro-analyses. David M. Berry, "The Computational Turn: Thinking about the Digital Humanities," *Culture Machine* 12 (2011): 1–22. Camille Roth, "Digital, Digitized, and Numerical Humanities," *Digital Scholarship in the Humanities* 34, no. 3 (2019): 616–32, https://doi.org/10.1093/llc/fqy057. On a more global history of alchemy, Alexandre M. Roberts. 2025. "Disentangling Alchemy." *Isis* 116 (4): 623–43, https://doi.org/10.1086/738213.

computational analysis are then further scrutinised, interpreted, and developed into narrative histories, by historians doing the work of traditional scholarship.

Indeed, we maintain that computational methods do not reduce the historians' engagement with their sources; they reconfigure it by focusing on aspects that only become interesting or significant because of the specific kinds of questions that digital manipulation and analysis invite, or sometimes require to ask. Indeed, errors introduced by computational techniques breed pairs of questions pertinent both to improving the code and to reconsidering how a document under consideration may have been conceived and constructed.[11] This process is elucidated in reference to the projects discussed in several articles in this special issue.[12]

We acknowledge that skepticism persists – as discussed by chemist and historian of chemistry Guillermo Restrepo, a co-editor of this volume, in his first contribution to this special issue.[13] Some historians remain wary of what they perceive as the "decontextualisation" of the past; others argue that computational methods privilege historical narratives at the macrolevel at the expense of history at the microlevel; others focus on the alleged fragility of the digital data.[14] These concerns are legitimate and they are precisely why this special issue is necessary: to demonstrate how computational approaches can be methodologically reflexive, historically sensitive, and epistemologically self-aware.

The framework for the computational history of chemistry presented in this special issue has been forged through years of engagement with historians, chemists, computer scientists, and philosophers across diverse forums. It reflects the collective experiences of the editors and practitioners who have witnessed both the reception of computational results and the questions, concerns, and opportunities they provoke in the historical community.

At its core, computational history is the disciplined, reflexive, and historically grounded application of mathematical and statistical techniques to analyse, model, simulate, or visualise historical data in order to uncover patterns, test hypotheses, and generate new understandings of the past. It begins with digital data – born from the digitisation, curation, and annotation of historical sources – but it is not merely about data processing. Rather, it is a form of historical reasoning,

11 For an example see Sarah Lang, et al., "Confabulated Transliterations? Managing the Lure of Plausibility in LLM-Detected Arabic Terms in an Early Modern Lexicon," in *Critical Approaches to Automated Text Recognition*, ed. Melissa Terras, et al. (London: Facet Publishing, forthcoming 2026).

12 See articles by Guillermo Restrepo, "Computational History of Chemistry – Part 1: Challenges And Opportunities;" and "Computational History of Chemistry – Part 2: Multiple Scales, from the Individual To the Global." See also Sarah Lang, Farzad Mahootian, Hazem Lashen, "Mediating Alchemical Language Across Terminologies and Cultures in Ruland's *Lexicon Alchemiae*. A Data-Driven Study of Arabic Terms"; Georgiana Hedesan, Petr Pavlas, and Vojtěch Kaše, "An Alchemical Prima Materia for the Digital Age: Making the EMLAP (Early Modern Latin Alchemical Prints) Dataset"; Sarah Lang, Petr Pavlas, Vojtěch Kaše, "Contextual Word Embeddings for Paracelsian Lexicography: Tangled Terminologies and their Origins in Ruland's Alchemical Dictionary." All: *Ambix*, this issue.

13 Guillermo Restrepo, "Computational History of Chemistry – Part 1," *Ambix*, this issue.

14 On decontextualisation, see Katherine Bode, "The Equivalence of 'Close' and 'Distant' Reading; or, Toward a New Object for Data-Rich Literary History," *Modern Language Quarterly* 78, no. 1 (2017): 77–106. On the tension between narratives at the macro and microlevels, see Benjamin Mangrum, "Aggregation, Public Criticism, and the History of Reading Big Data," *PMLA* 133, no. 5 (2018): 1207–24. On data fragility, see Jeff Rothenberg, "Ensuring the Longevity of Digital Documents," *Scientific American* 272, no. 1 (1995): 42–47.

where computational algorithms serve as tools for discovery. Crucially, the historian remains central at every stage: from the selection and curation of sources, to the design of data structures, the choice of analytical methods, and the interpretation of results. The computer scientist contributes expertise in modelling and implementation, but the historian provides the interpretive framework, the contextual sensitivity, and the critical judgment that situates data at every phase of the research process, including iterative curation, annotation, analysis, reanalysis, re-annotation, etc. The historian's role is foundational in shaping both what is said and what is asked. In this way, computational history amplifies the historian's capacity to ask questions and to analyse sources in new ways that will occasionally stimulate new interpretations of established narratives.

As this brief overview illustrates, the computational history of chemistry is inherently interdisciplinary, drawing on both the practices of historical scholarship and the methods of computer science. It rests on a foundation of digital data – born from curation, digitisation, and annotation – while relying on algorithms to process and analyse these materials, and on computational techniques to model, simulate, and visualise historical patterns. The integration of artificial intelligence further expands the field's reach, enabling new forms of pattern detection, hypothesis generation, and large-scale analysis.

Yet this convergence of disciplines demands precision. The terms *digital*, *algorithmic*, *computational*, and *AI* are often used interchangeably, but they denote distinct concepts with different epistemological implications.[15] Misuse or conflation of these terms risks obscuring the nature of the work, reinforcing misconceptions, and alienating scholars trained in traditional historiography. To engage meaningfully with the field, it is essential to clarify these distinctions in order to establish a shared language that enables rigorous dialogue. Only with such clarity can we begin to appreciate the unique roles each approach plays, their methodological foundations, and their relationship to established historical practice.

Digital refers to the representation, storage, processing, and transmission of information in discrete, quantifiable units – binary code (0s and 1s) – using electronic systems. In historical scholarship, a digital approach involves converting analogue sources (handwritten manuscripts, printed books, laboratory notebooks, or physical artifacts) into machine-readable formats through processes such as digitisation, optical character recognition (OCR), metadata tagging, and database creation. The digital form enables new modes of access, preservation, and analysis, and it may, depending on the researcher's degree of familiarity with it, involve interpretive decision making to a greater or lesser extent. A digital archive is not necessarily a raw repository of data, it is the cumulative result of expert decisions great and small and a precondition for further work.[16]

[15] Some of these terms are discussed in detail in Restrepo, "Computational History of Chemistry - Part 1."

[16] The computational revolution began after World War II and has remained digital ever since. However, contemporary advances in computer sciences make it possible to foresee a further revolution, one that frees the computational enterprise from the binary trap of 0s and 1s and opens multiple possibilities through the advent of qubits and

This crucial step in the computational history of chemistry is vividly illustrated in the article by historian of science Georgiana Hedesan, philosopher of science Petr Pavlas, and social scientist Vojtěch Kaše, whose meticulous approach reveals that digitisation is far more than the mechanical conversion of analogue sources, such as early modern alchemical manuscripts in Latin, into binary files.[17] Their project decisively overturns the misconception that digitisation is merely about creating images of old documents. If it were, the computational history of chemistry would have begun decades ago, with the advent of the digital camera. Instead, their work demonstrates that digitisation is an active, interpretive practice – one in which the historian is not a passive observer but a skilled agent wielding digital tools. It involves deliberate decisions: which documents to prioritise, which algorithms to select, and how to tune and adapt them to capture historically significant content, from the nuances of alchemical Latin to the meanings of symbolic notations. It also requires careful judgment of how to represent the document's structure (how to preserve marginalia, layout features, and ornamental elements) maintaining fidelity to the original while providing access to meaningful historical information. Hence, digitisation is both an act of preservation and a foundational moment of historical interpretation, where the choices made shape what can be known in any given analytical process.

A further term often invoked in computational history is *algorithmic*, which describes a systematic, rule-based procedure for solving a problem or performing a task, typically implemented in software. An algorithm is a finite sequence of well-defined steps that takes input data and produces output according to a set of instructions. In historical research, algorithmic methods automate processes such as text classification, entity recognition, pattern detection, or network analysis. Yet algorithms are far from neutral tools.[18] In the particular case of the algorithms employed in the works presented in this special issue, they are deeply embedded in interpretive decisions – each one a reflection of the historian's conceptual framework, methodological choices, and epistemic commitments.

The scripts used across these articles span a wide range of functions: from detecting marginalia in alchemical texts or extracting alchemical, medicinal, and astrological symbols,[19] to generating high-dimensional semantic spaces that capture the relationships between words in historical corpora.[20] Others convert contemporary chemical formulae into forms consistent with pre-1860 atomic weight systems – requiring careful approximation of fractions into chemically meaningful

[16] *Continued*
computing devices able to handle information stored in this amplified fashion. See Eleanor G. Rieffel and Wolfgang H. Polak, *Quantum Computing: A Gentle Introduction* (Cambridge, MA: MIT Press, 2014).

[17] Georgiana Hedesan, Petr Pavlas, and Vojtěch Kaše, "An Alchemical Prima Materia for the Digital Age: Making the EMLAP (Early Modern Latin Alchemical Prints) Dataset".

[18] Geoffrey C. Bowker and Susan Leigh Star, *Sorting Things Out: Classification and Its Consequences* (Cambridge, MA: MIT Press, 2000).

[19] Hedesan, Pavlas, and Kaše, "An Alchemical Prima Materia for the Digital Age."

[20] Sarah Lang, Petr Pavlas, Vojtěch Kaše, "Contextual Word Embeddings for Paracelsian Lexicography: Tangled Terminologies and their Origins in Ruland's Alchemical Dictionary."

integers.[21] Some transform chemical data into networks of reactions, while others identify clusters of chemical elements based on the compounds they form.[22] Algorithms of computational linguistic studies discussed in this issue measure the variation of information content across diachronic corpora, analyse nomination networks for the Nobel Prize in Chemistry, and map citation patterns to trace the influence of individual scientists. Still others merge bibliographic and chemical data to disentangle the role of nations in the global production of chemical knowledge.[23]

The human in the loop: computational history as a practice of interpretation

Each of these algorithms embodies a series of interpretive choices. What counts as an alchemical symbol, and how is it distinguished from an astrological reference? Which mathematical space best captures semantic similarity in a corpus – word2vec (short for Word to Vector, an algorithm for representing words as numerical vectors in a continuous space) BERT (which stands for Bidirectional Encoder Representations from Transformers, Google's advanced deep learning model), or a custom-built embedding model? What is the most appropriate method for approximating atomic weights in a way that preserves chemical plausibility? Should chemical reaction networks be modelled as graphs or hypergraphs to account for multi-reactant, multi-product transformations? Which measure of information content best reflects meaningful shifts in discourse over time? These are not technical details – they involve the historian reasoning about the nature of the data at hand, about the historical question to solve or the historical hypothesis to test:[24] The algorithm is not a black box; it is a site of interpretation. The historian does not simply apply a tool – they design it, calibrate it, and justify it. In this way, the algorithm becomes but an active participant in the historical inquiry.

Computational is a further term used in computational approaches to the history of chemistry, denoting the application of mathematical, statistical, or algorithmic techniques to analyse, model, or visualise datasets. What makes it computational – distinct from digital or algorithmic – is its active role in inquiry. While digital is about format and algorithmic is about process, computational is about discovery. It is an analytical tool for stimulating insight. Based on the digital record and

[21] Section titled "Emergence and post-1860 development of the periodic system" in Guillermo Restrepo, "Computational History of Chemistry – Part 2: Multiple scales, from the individual to the global.".

[22] Section titled "Large-scale patterns and major transitions in the unfolding of the material outcome of chemistry" of Restrepo, "Computational History of Chemistry – Part 2."

[23] On the information content of diachronic corpora, nomination networks, citation patterns, and national contribution to chemical discovery, see Restrepo "Computational History of Chemistry – Part 2," sections titled "Approaches to the history of chemistry from a computational linguistic perspective," "The politics of excellence: the Nobel Prize in Chemistry,' 'Biography and scientific careers," and "Geopolitics of substance discovery and the global reconfiguration of chemical practice," respectively.

[24] These decisions are particularly evident in Hedesan, Pavlas, and Kaše "An Alchemical Prima Materia for the Digital Age," where the reader is not only confronted with the choices made during digitisation but also with the enduring tension in computational scholarship: balancing computational efficiency with historical accuracy and comprehensiveness.

algorithmic procedures, computational methods help historians to develop historical narratives. At the intersection of data, algorithm, and interpretation, computational history is a form of historical reasoning.

For instance, computational analysis can reveal structural transitions in history, simulate the diffusion of terminology, or reconstruct the evolution of the theoretical frameworks from historical data.[25] Computational tools enable historians to move beyond descriptive analysis into the realm of generative inquiry. They can detect latent semantic structures in alchemical manuscripts, or flag anomalies in citation networks.[26] Most importantly, they support hypothesis generation and counterfactual exploration – a transformative capability.[27]

Take, for example, the case discussed by Restrepo: the analysis of the pace of reported new chemicals from 1800 to the present.[28] The study revealed a remarkably stable exponential growth, unaffected even by global disruptions such as World Wars. This raises a fundamental historical question on the structural or epistemic forces that sustain such a consistent rate of material innovation across centuries and crises. Similarly, computational linguistic analysis of English corpora showed that the eighteenth-century chemist, George Pearson (1751–1828), played a pivotal role in spreading Lavoisian terminology in England.[29] This prompts a deeper inquiry: why was Pearson so central to this diffusion, and did similar patterns of conceptual transmission occur in other languages and regions? Such questions point to the geopolitical and social dynamics of scientific change.

At the conceptual level, computational tools have revealed a striking stability in chemistry's core vocabulary, in contrast to the more relaxed structures of disciplines like botany or cosmology.[30] This invites critical questionning on what this stability implies about the epistemic foundations of chemistry, and how they emerged from the interplay of material practice, theoretical innovation, and linguistic standardisation. Still at the conceptual level, computational linguistic studies have shown that the conceptual structure of chemistry changed little from the second half of the eighteenth century (when phlogiston theory was the leading chemical theory) to the period after the so-called Lavoisian chemical revolution.[31] This clearly shows the contribution of the computational history of chemistry to current debates in the history and philosophy of chemistry.[32]

25 These cases are discussed in detail in Restrepo, "Computational History of Chemistry – Part 2.".
26 See Lang, "Contextual Word Embeddings," and Restrepo, "Computational History of Chemistry – Part 2."
27 On counterfactual analysis carried out by using computational tools, see section titled "Emergence and post-1860 development of the periodic system," in Restrepo, "Computational History of Chemistry – Part 2."
28 Section titled "Large-scale patterns and major transitions in the unfolding of the material outcome of chemistry" in Restrepo, "Computational History of Chemistry – Part 2."
29 Section titled "Approaches to the history of chemistry from a computational linguistic perspective," in Restrepo "Computational History of Chemistry – Part 2."
30 Section titled "Approaches to the history of chemistry from a computational linguistic perspective," in Restrepo "Computational History of Chemistry – Part 2.".
31 Restrepo "Computational History of Chemistry – Part 1," and "Computational History of Chemistry – Part 2.".
32 In this respect, for instance, philosopher of science Hasok Chang, has championed the idea of a disruptive chemical revolution at the conceptual level by suggesting that the phlogiston theory already encoded the conceptual structure of the Lavoisian chemistry, see Hasok Chang, *Is Water H2O?: Evidence, Realism and Pluralism*, vol. 293, Boston

Turning to alchemy, historian, and co-editor of this volume, Sarah Lang and social scientist Vojtěch Kaše's article on Ruland's *Lexicon Alchemiae* – which maps its lexical footprint across pre-1612 alchemical prints – reveals that the lexicon was not a singular creation, but a compilation drawing from prior textual authorities, with strong affinities to Libavius's *Alchemia* and Dorn's *Dictionarium*.[33] This raises a profound question: How did reference works like Ruland's function? Did their role go beyond just repositories of knowledge, as they have long been thought of, being active agents in reshaping it? In a further study by Lang, along with Farzad Mahootian, and computer scientist Hazem Lashen, large language models are used to detect Arabic-derived terms in the lexicon.[34] The study revealed that Arabic terms were not simply imported, but fragmented, dispersed, and recontextualised across multiple entries under variant spellings. This leads to critical questions around the role that non-Western knowledge played in Latinate early modern alchemy. Did scholars actively engage with Arabic texts, attempting to draw directly from primary sources? Or did they, due to a lack of understanding of the Arabic language, find different alchemical *Decknamen* in transmission variants that were, in fact, not meaningful differences in the source texts?

Likewise, computational tools have opened the door to counterfactual exploration. Restrepo's analysis of alternative periodic systems, based on pre-1860 atomic weights, reveals that the structure of the periodic system was already latent in the collective of chemicals discovered decades before the formal articulation of the system.[35] This prompts a transformative question: What role did epistemic, social, and institutional factors (beyond the material record) play in the eventual emergence of the system in the 1860s?

Finally, after digital, algorithmic, and computational (each of which has transformed how historians engage with sources) there is a term that today very often comes to mind when discussing the role of computers in the history of chemistry, and arguably in any human endeavour: *Artificial Intelligence (AI)*. This branch of computer sciences constitutes a powerful extension of the discussed computational approach – one that amplifies our ability to detect patterns, explore complexity, and generate new questions.

To understand what AI is, consider a large language model (LLM) – a type of AI trained on vast amounts of text, such as books, articles, and manuscripts. It learns to predict the next word in a sentence by analysing billions of examples. However, it does not "understand" meaning in the way a human does. Instead, an LLM

[32] *Continued*
Studies in the Philosophy and History of Science (Dordrecht: Springer Netherlands, 2012) 1–70. As argued by Restrepo, this seems to be confirmed by the computational linguistic studies discussed in his contributions to this special issue, see Restrepo "Computational History of Chemistry – Part 1," and "Computational History of Chemistry – Part 2."

[33] Kaše and Lang, "Contextual Word Embeddings."

[34] Lang, Mahootian, Lashen, "Mediating Alchemical Language Across Terminologies and Cultures in Ruland's *Lexicon Alchemiae*.".

[35] Section titled "Emergence and post-1860 development of the periodic system" of Restrepo, "Computational History of Chemistry – Part 2."

identifies statistical patterns – how words tend to cluster, and how phrases relate. This is how it can generate text, answer questions, or even suggest new interpretations.[36]

A clear example from this special issue illustrates this process. In the study of Ruland's *Lexicon Alchemiae*, Lang and Kaše used word embeddings – a technique where each word is represented as a point in a high-dimensional space based on its surrounding context.[37] For instance, the term *oleum sulphuris* appears in different texts with varying meanings.[38] The AI model captures these nuances by placing similar usages close together in the space, while distinguishing them from others. This allows historians to see, at a glance, how a term was used across time and across authors in hundreds of instances – the replication of which, through close reading alone, would be prohibitively time-consuming.

This is the strength of AI: it can process vast, complex data (millions of words, hundreds of manuscripts) and reveal patterns that are invisible to the human eye. It can detect subtle shifts in terminology, identify hidden networks of influence, or flag anomalies in citation patterns.[39] In the history of chemistry, this means that we can trace the spread of ideas across the globe, or map the rise of a particular innovation in chemical discovery with unprecedented precision.

Yet AI, in many ways, is inherently flawed.[40] It does not know what is true or false. It cannot distinguish between a reliable source and a forgery. It may "hallucinate" – generate plausible-sounding but incorrect information. Moreover, as it relies on information gathered by humans throughout history, its output is bound to repeat the inherent biases involved in the selection, curation, and classification of sources.[41] AI is trained on data with a very high proportion of western, and male-authored records, depicting institutional biases and other sorts of data inequalities, where non-European, non-male, and non-conventional institutional voices are most often missing. In the case of alchemical texts, an AI might misidentify a term as Arabic because of a spelling variant, or miss a term entirely for a variety of reasons.[42]

[36] On the technical details of AI, including LLMs, without going too deep on the mathematics involved, see Anil Ananthaswamy, *Why Machines Learn: The Elegant Math Behind Modern AI* (New York, NY: Penguin Group, 2025). A more historically oriented account of the unfolding of AI, with brief explanations of the main algorithms of the field, is found in Brian Christian, *The Alignment Problem: How Can Machines Learn Human Values?* (London: Atlantic Books, 2021).

[37] Some relevant references on word embeddings include Yoshua Bengio et al., "A Neural Probabilistic Language Model," *Journal of Machine Learning Research* 3 (2003): 1209–27; Tomas Mikolov et al., "Distributed Representations of Words and Phrases and Their Compositionality," *Proceedings of the 27th International Conference on Neural Information Processing Systems - Volume 2 NIPS*'13, no. 2 (2013): 3111–19; Tomas Mikolov et al., "Efficient Estimation of Word Representations in Vector Space," arXiv:1301.3781, preprint, arXiv, 7 September 2013, https://doi.org/10.48550/arXiv.1301.3781.

[38] Lang, "Contextual Word Embeddings."

[39] Christian, *The Alignment Problem*.

[40] For a discussion of critical concerns regarding the use of Large Language Models (LLMs) in the history of science (that apply to many other types of AI, too), see: Sarah Lang, "Critical Concerns for Using LLMs."

[41] Christian, *The Alignment Problem*.

[42] The indispensable role of the historian in guiding and correcting the computational process is vividly illustrated in Hedesan, Pavlas, and Kaše, "An Alchemical Prima Materia for the Digital Age," where the authors detail the

This is why the human historian remains essential – despite marketing dreams of automating the hermeneutics and the practice of history, as discussed in Restrepo's detailed treatment of the so-called Nobel Turing test and its implications for the practice of history.[43]

Because of the aforementioned shortcomings of the application of AI, the historian does not step aside. Instead, they guide the process: choosing which data to use, designing the questions, interpreting the outputs, and questioning the assumptions. In the Ruland *Lexicon* study, for example, AI flagged hundreds of potential Arabic terms – but only human experts could verify them, distinguish between real and false positives, and interpret their significance.[44] The AI did the heavy lifting of scanning; historians and linguists did the critical work of assigning meaning to the results.

To a far greater extent than earlier analogue tools, AI models and the infrastructure on which they depend are built upon forms of extraction that affect our natural resources and, more importantly, rely upon the labour of, often underacknowledged, data workers operating under unacceptable conditions.[45] It is, however, necessary to distinguish between the technology itself and the circumstances under which it is deployed: the same model may be provided by a company whose labour practices cause lasting psychological harm or exploit local communities, while the technology could, at least in principle, be operated under more ethical conditions. Nevertheless, the training data used by most popular models remain problematic.[46] Beyond issues of bias and data quality, these datasets frequently contain pirated or otherwise non-consensually obtained material.[47] Furthermore, particularly in the case of image-generation systems, the development of both models and datasets is often entangled with military and surveillance contexts.[48]

In this light, we argue that it is a misunderstanding to think of AI as a threat to historians and historiographers. Instead, AI could be thought of as a new kind of instrument – one that, like the microscope or the telescope, extends human capacity to see and to ask new questions without replacing the eye, the mind, or the faculty of judgment. This analogy, however, has some limitations. In addition to extending the range of accessible data, AI and other computational techniques also draw

42 *Continued*
limitations of AI-driven text and image recognition, and demonstrate how targeted human intervention (through manual curation and algorithmic refinement) was essential to overcoming these shortcomings.

43 Restrepo, "Computational History of Chemistry - Part 1.".

44 Lang, "Contextual Word Embeddings."

45 For more information see Sarah Lang, "Critical Concerns for Using LLMs."

46 Amandalynne Paullada, et al., "Data and its (Dis)Contents: A Survey of Dataset Development and Use in Machine Learning Research," *Patterns* 2 (2021): 1–14.

47 Jathan Sadowski, "When Data is Capital: Datafication, Accumulation, and Extraction," *Big Data and Society* 6 no. 1 (2019): 1–12; Michelle Rademeyer and Niloufer Selvadurai, "Out from the Shadows: Developing Effective Copyright Laws for AI Training Datasets and Shadow Libraries," *Journal of Intellectual Property Law & Practice* 21, no. 1 (2026): 22–35.

48 Lucy Suchman, "Algorithmic Warfare and the Reinvention of Accuracy," *Critical Studies on Security*, 8, no. 2 (2020): 175–87; Pieter Verdegem, "Dismantling AI Capitalism: The Commons as an Alternative to the Power Concentration of Big Tech," *AI & Society* 39 (2024): 727–37; Pratyusha Ria Kalluri, et al., "Computer-Vision Research Powers Surveillance Technology," *Nature* 643 (2025): 73–79.

inferences about data, whether that data is wavelengths and intensities of light through a telescope, or letters on a page of a partially legible, twelfth century manuscript. Furthermore, like any instrument, AI introduces artifacts that are specific to the instrument's interaction with its environment, its tolerance limits, and so on. Early telescopes, for example, limited astronomy to visible light; and astronomy today employs a variety of kinds of telescopes, each with its clearly articulated, known limits. In chemistry, this knowledge of instrumental potentialities, but also of their limits and artifacts, is included in courses on instrumental analysis and is a required part of chemical education and laboratory protocol. However, while model evaluation and diagnostic benchmarking might be considered functional analogues to instrumental analysis, arguably their efficacy and reliability are severely limited to the point that the analogy breaks down.[49] We believe that this important difference should not dissuade computational historians from considering AI as an instrument, but it makes "human in the loop" modality an indispensable part of any AI-assisted research project.

Situating the field

Computation is reshaping knowledge across disciplines, fostering new hybrid approaches and interdisciplinary communities.[50] While these communities often retain ties to their disciplinary roots, they also evolve in response to shared methodological challenges and technological advances. In this dynamic landscape, the computational history of chemistry occupies a unique position: it is both rooted in the core traditions of the history of chemistry and engaged with two emerging scholarly fields – digital/computational history and digital/computational humanities.[51] This section explores how the computational history of chemistry navigates these intersections, drawing on shared tools and epistemologies while maintaining its distinct historical focus.

[49] Jonathan Kim, et al., "Limitations of Large Language Models in Clinical Problem-Solving Arising from Inflexible Reasoning," *Scientific reports* 15, no. 1 (2025): 39426.

[50] Anthony J. G. Hey et al., *The Fourth Paradigm: Data-Intensive Scientific Discovery* (Redmond, WA: Microsoft Research, 2009); Urszula Pawlicka-Deger, "Infrastructuring Digital Humanities: On Relational Infrastructure and Global Reconfiguration of the Field," *Digital Scholarship in the Humanities* 37, no. 2 (2022): 534–50, https://doi.org/10.1093/llc/fqab086.

[51] As noted above, there are important methodological and disciplinary distinctions between the terms *digital* and *computational*. The details are too complex to discuss fully here, but it is important to note that these terms are not used merely as synonyms within the relevant scholarly traditions. Practitioners of these fields generally expect them to be employed with some degree of precision. In both *digital history* and *digital humanities*, *digital* is usually understood as the broader umbrella term. It also predates what is generally described as the "computational turn" in the development of these fields and can therefore be used in a more general sense to refer to research practices that engage with and operate on the basis of digital data. By contrast, *computational* is typically reserved for more specific methods and subfields that rely on quantitative, numerical or algorithmic analysis. While all computational approaches can safely be referred to as digital humanities methods, not all digital approaches are necessarily computational in this narrower sense. See Berry, "The Computational Turn," 1–22; Roth, "Digital, Digitized, and Numerical Humanities," 616–32. Note that while Digital Humanities and Computational Humanities refer to the use of digital or computational methods in the Humanities overall, there are differences in how various disciplines, including history, have adopted digital methods. Differences in disciplinary approaches are discussed in more detail in Sarah Lang, "Epilogue: A Computational Turn? Reflections on Digital and Computational History of Science, Knowledge and Technology," *Ambix,* this issue.

The computational history of alchemy and chemistry must ultimately be situated within the broader domain of digital history, an area that has gained traction since the early 2000s, particularly through the work of historians Daniel J. Cohen and Roy Rosenzweig.[52] Unlike earlier forms of quantitative history, which focused on statistical analysis of historical data, we suggest that digital history represents a methodological shift: it is not merely about numbers, but about structured reasoning, algorithmic processing, and the generation of insight from complex datasets. This distinction is crucial. As Cohen and Rosenzweig argued, computational history enables historians to move beyond descriptive analysis into the realm of generative inquiry – where patterns are observed, modelled, simulated, and tested.

Social scientist Camille Roth distinguishes between digital humanities which encompasses methods carried out in the digital realm but remains hermeneutically and epistemologically aligned with traditional scholarship, and computational humanities, which introduces new modes of reasoning that go beyond established historical practice.[53] This distinction mirrors the growing divide between *digital history* and *computational history*: the latter, by analogy with computational humanities, now describes research that is methodologically, epistemologically, and hermeneutically distinct from traditional approaches – much like experimental methods in the history of science, which generate new perspectives on historical data.

Until recently, the histories of alchemy and chemistry have remained relatively underexplored in the context of computational humanities. While experimental reconstructions of alchemical recipes, such as those by the historians of alchemy Lawrence Principe and William Newman, and projects like *Making and Knowing* have demonstrated the power of "practical exegesis," and digital scholarly editing has begun to address the complexities of alchemical manuscripts, the full potential of computational analysis has only recently been tapped.[54] The field is now poised to move beyond descriptive cataloging toward generative inquiry. Using machine learning, network analysis, computational linguistics, and AI-driven pattern detection, researchers can model the evolution of alchemy and chemistry, the dynamics

52 Daniel J. Cohen and Roy Rosenzweig, "Digital History: A Guide to Gathering, Preserving, and Presenting the Past on the Web" (Philadelphia, PA: University of Pennsylvania Press, 2005).

53 Roth, "Digital, Digitized, and Numerical Humanities."

54 On the term "practical exegesis," see Jennifer M. Rampling, *The Experimental Fire: Inventing English Alchemy, 1300–1700*, Synthesis (Chicago, IL: University of Chicago Press, 2023): 63–64, 97–99, 354. On performative methods more generally, see Hjalmar Fors et al., "From the Library to the Laboratory and Back Again: Experiment as a Tool for Historians of Science," *Ambix* 63, no. 2 (2016): 85–97; Marieke M. A. Hendriksen, "Rethinking Performative Methods in the History of Science," *Berichte Zur Wissenschaftsgeschichte* 43, no. 3 (2020): 313–22; Sven Dupré, et al. eds, *Reconstruction, Replication and Re-enactment in the Humanities and Social Sciences* et al. (Amsterdam: Amsterdam University Press: 2020); Lawrence Principe, *The Secrets of Alchemy* (Chicago, IL: University of Chicago Press, 2013). An example for an early digital scholarly edition is William R. Newman, ed, *The Chymistry of Isaac Newton* (Bloomington: Indiana University, 2005–2026), http://chymistry.org (accessed 3 May 2026). Pamela H. Smith, et al. eds, *The Making and Knowing Project: Intersections of Craft Making and Scientific Knowing* (2020), https://www.makingandknowing.org/bnf-ms-fr-640/ (accessed 3 May 2026), contextualised in: Sarah Lang, "Ways of Making Digital Editions of Historical Recipes and Experimental Texts: The Making and Knowing Project," *RIDE: A Review Journal for Scholarly Digital Editions and Resources* 21 (2026), https://ride.i-d-e.de/issues/issue-21/makingandknowing/.

of communities of research and practice, and the semiotic transformations that underlay historical changes of theory and practice.

The aforementioned evolution toward generative inquiry is particularly evident in the computational history of science, a subfield championed by historians such as Manfred Laubichler and Jürgen Renn.[55] Its roots stretch back to the mid-twentieth century, when historians began applying statistical methods to large-scale questions – such as population trends, mortality patterns, and the structure of scientific reputation.[56] These early efforts, part of the "quantitative history" movement of the 1960s, laid the groundwork for treating historical data as analysable, patterned, and meaningful. The field gained further momentum with the 2019 special issue of *Isis* devoted to computational history of science, and with the establishment of a dedicated research group at the Max Planck Institute for the History of Science (2018–2020).[57] Yet, despite this growing visibility, early computational approaches in the history of science were often geographically and temporally narrow – focused on twentieth-century developments in physics and biomedical sciences, with chemistry playing a marginal role.[58]

As Lang notes in the epilogue of this special issue, what has been referred to as "the computational history of science" was often just an umbrella term that described computational work on the history of a particular branch of science, such as the life sciences, and thus, has initially overlooked chemistry – despite its central role in the development of modern science.[59] This gap presents a unique opportunity: the early stages of the computational history of chemistry can learn from the achievements and pitfalls of earlier computational work in other sciences, avoiding past limitations while building on proven methodologies. But it is imperative to acknowledge limits on the transferability of methods due the specificity of any computational research programme necessitated by the peculiarities of the subject matter (use of symbolism, terminology, etc.), and by the continual update cycles endemic to the rapid pace of computer technology.[60]

[55] Laubichler et al., "Computational Perspectives in the History of Science"; Laubichler et al., "Introduction"; Jürgen Jost et al., "Computational History: Challenges and Opportunities of Formal Approaches," *Journal of Social Computing* 4, no. 3 (2023): 232–42; Manfred Laubichler et al., "Computational History of Knowledge: Challenges and Opportunities," *Isis* 110, no. 3 (2019): 502–12. On computational history, see Cohen and Rosenzweig, *Digital History*; Lang, "(Doing) Computational History."

[56] François Furet, "Quantitative History," *Daedalus* 100, no. 1 (1971): 151–67; Douglass C. North, 'The New Economic History After Twenty Years', *American Behavioral Scientist* 21, no. 2 (1977): 187–200; Steven Ruggles, "The Revival of Quantification: Reflections on Old New Histories," *Social Science History* 45, no. 1 (2021): 1–25; Eugene Garfield, Irving H. Sher, and Richard J. Torpie, *The Use of Citation Data in Writing the History of Science*. (Philadelphia, PA: Institute for Scientific Information, 1964), https://garfield.library.upenn.edu/papers/useofcitdatawritinghistofsci.pdf; Derek John de Solla Price, *Little Science, Big Science* (New York, NY: Columbia University Press, 1963).

[57] "Computational History of Science," MPIWG, https://www.mpiwg-berlin.mpg.de/computational-history (accessed 21 May 2026).

[58] Roberto Lalli et al., "The Socio-Epistemic Networks of General Relativity, 1925–1970," 2020; Laubichler et al., "Computational History of Knowledge."

[59] Lang, "Epilogue."

[60] Damerow and Wintergrün, "The Hitchhiker's Guide," 514; Jonathan Kim, Anna Podlasek, et al. "Limitations of Large Language Models."

This special issue is situated at the convergence of three intellectual traditions: the history and philosophy of science, the long-standing practice of quantitative history, and the more recent turn toward computational and digital methods in the humanities. From early advocates of quantification, such as historian William O. Aydelotte, who stated that "counting, when circumstances permit it, may assist in the explanation of a limited class of historical problems," to reflections by physicist and scientometrician, Derek de Solla Price, who revealed regularities in the growth and structure of scientific activity, quantitative approaches have repeatedly demonstrated their heuristic value.[61]

Historian of chemistry Jeffrey A. Johnson's contribution to this special issue provides a vital historical bridge between these traditions.[62] His article traces the emergence of quantitative approaches to the history of chemistry from the 1960s and 1970s onward, showing how early efforts – though limited by the technology of the time – were already computational in spirit. Johnson's work is not just a retrospective account; it is a first-hand witnessing of a transformation starting in the 1960s and 1970s, when historians began to use punched cards, mainframe computers, and early databases to study the careers of chemists, and the structure of scientific communities. His narrative reveals that the seeds of computational history were sown long before the digital age – when scholars such as historian of science Arnold Thackray, and sociologist of science Joseph Ben-David began to model scientific change using statistical tools.[63]

We argue that this continuity is essential. Indeed, the emergence of computational history of chemistry is anything but a sudden innovation or takeover by outsiders; we hold that it is the natural evolution of a long-standing historiographical tradition. While the use of computational methods in the history of chemistry began as a rather niche endeavour, the emergence and widespread proliferation of modern digital tools has amplified its reach considerably. Johnson's work reminds us that the field has a long history.

At the methodological level, there is strong resonance between the computational history of chemistry and the digital humanities. Both rely on computational tools to process large repositories of textual data. Yet the digital humanities have also pioneered the analysis of images, methods which are now entering the realm of chemical history, as seen in the detection of alchemical symbols in Hedesan, Pavlas, and Kaše's contribution.[64]

Despite the advantages computational methods bring to the practice of the history of chemistry, the field is not without challenges and criticisms. As discussed

[61] William O. Aydelotte, "Quantification in History," *The American Historical Review* 71, no. 3 (1966): 803–25; Price, *Little Science, Big Science*.

[62] Jeffrey A. Johnson, "Emergence of New Quantitative Approaches to the History of Chemistry after 1950," *Ambix*, this issue.

[63] Steven Shapin and Arnold Thackray, "Prosopography as a Research Tool in History of Science: The British Scientific Community 1700–1900," *History of Science* 12, no. 1 (1974): 1–28; Joseph Ben-David, "Scientific Productivity and Academic Organization in Nineteenth Century Medicine," *American Sociological Review* 25, no. 6 (1960): 828–43.

[64] Hedesan, Pavlas, and Kaše, "An Alchemical Prima Materia for the Digital Age.".

by Restrepo, these are inherited from the history of science and the broader practice of history. They reflect enduring tensions in historical inquiry: the risk of decontextualization, the tension between micro and macrolevels of detail, the fragility of data, and the opacity of algorithms.[65] We address these concerns in the next section as opportunities to reengage with questions of the discipline's methodological rigour and epistemic responsibility.

Reclaiming the historian's responsibility

As briefly discussed earlier, the computational history of chemistry has faced a range of criticisms observed in our presentation of results in the field and in other interactions with historian colleagues, such as peer review processes. These criticisms are thoroughly addressed in Restrepo "Computational History of Chemistry – Part 1," but here, we briefly revisit those that we regard as most relevant.

The most pressing challenges of the computational history of chemistry range from accusations of decontextualisation and the overemphasis of macrolevel studies at the expense of historical studies at the microlevel to concerns about the fragility of digital data.[66] These concerns are not merely technical; they touch on fundamental questions about what it means to know the past. Yet rather than signalling a failure of the method, these challenges often reflect enduring tensions in historical practice that have shaped the discipline.[67] The real question is not whether computational history is vulnerable to these issues, but whether it can respond to them with the same or greater transparency, reflexivity, and rigour as traditional approaches do.

The fear that computational history decontextualises the past – reducing alchemical manuscripts to word counts, or the discovery of chemicals to numbers – is understandable. Yet, paradoxically the very act of building a computational project demands rigorous engagement with context, just as in traditional close reading. However, with computational history, attention is directed to the crucial interface between text and computer code. To train an algorithm to recognise "mercury" in a sixteenth-century manuscript is not to simplify the text, it is to confront its complexity: the shifting meanings of the term across languages, traditions, and symbolic systems; the variations in spelling; the layers of metaphor; the hidden references. This is not a mechanical task; it is a hermeneutic one, demanding that the historian define what counts as "mercury" in a given context – something that would be impossible to do well without deep philological and historical knowledge. Hedesan, Pavlas,and Kaše's contribution on the digitisation of early modern alchemical prints demonstrates this vividly.[68]

[65] These and several other challenges of the computational history of chemistry are discussed in depth in Restrepo, "Computational History of Chemistry – Part 1."

[66] See n. 4 for references on these criticisms.

[67] Zachary Schrag, *The Princeton Guide to Historical Research* (Princeton, NJ: Princeton University Press, 2021).

[68] Hedesan, Pavlas, and Kaše, "An Alchemical Prima Materia for the Digital Age."

This process forces historians to ask not just what a term means, but what it could mean in a different time, place, or tradition. And when these annotated texts are linked to institutional affiliations, geographic distributions, or material availability, the past is not flattened – instead its context is expanded, revealing the social, economic, and political forces that shaped it. This is evident in Lang, Pavlas, and Kaše's analysis of Ruland's *Lexicon Alchemiae*, where the dictionary emerges not as a neutral reference tool, but as a convergence point of multiple traditions – Paracelsian, alchemical, and even older classificatory systems like those of Abū Bakr al-Rāzī (Rhazes, 864/865–925/935 CE).[69] The lexicon's structure reveals not a single source, but a layered, hybrid knowledge system where meaning is negotiated rather than taken for granted.

This deeper contextualisation is also evident in Restrepo's analysis of the recent unfolding of chemical discovery.[70] While the global pace of substance discovery has remained remarkably stable, unaffected even by geopolitical tensions or economic crises, closer examination reveals dramatic shifts at the national level. The decline of the US in chemical innovation since 2013, for instance, is not a sign of stagnation, but a reconfiguration of global leadership, driven by the rise of institutions like the State Key Laboratory of Organometallic Chemistry in China, which has become a leading engine of organometallic chemistry. This shows that computational history does not erase context – it can reveal it at a scale, depth, and complexity previously unimaginable.

The criticism that computational history privileges macro-level narratives at the expense of micro-level inquiry rests on a false dichotomy – one that treats scale as a hierarchy rather than a methodological lens. In truth, the history of chemistry has always been shaped by both the individual and the system, the local and the global.[71] Computational tools do not replace the historian's focus on a single alchemist, a single experiment, or a single manuscript but can enable or enrich it. Under the right circumstances, they can sometimes even help recover what had hitherto been overlooked: a chemist whose work was buried in the academic records, or whose contributions to the spread of ideas were overlooked, a laboratory whose contributions have remained unnoticed by institutional histories.[72] By mapping

69 Lang, "Contextual Word Embeddings."

70 As discussed in Restrepo, "Computational History of Chemistry – Part 1," and "Part 2." The primary source for the study of the recent historical unfolding of chemical discovery is Marisol Bermúdez-Montaña, Angel Garcia-Chung, Peter F. Stadler, Jürgen Jost, and Guillermo Restrepo, "Global Shifts in the Geopolitics of Chemical Discovery: China's Rise and the Decline of US Influence," *Quantitative Science Studies* 7 (2026): 1–20.

71 This is not only true for the history of chemistry but for history in general, the result of the tension between determinism and contingency, as noticed by Tolstoy and later by Bloch. See Leo Tolstoy, *War and Peace* (Moscow: T. Ris, 1869, translated into Spanish as Guerra y paz, translators: Irene y Laura Andresco (Madrid: Alianza editorial, 2018): 855–911, and Marc Bloch, *The Historian's Craft* (Paris: Armand Colin, 1949, repr. Manchester: Manchester University Press, 1992): 171–186.

72 A case in point is the overlooked chemist brought to the fore by computational tools, George Pearson, who turned out to be a central figure in the spread of Lavoisian ideas in England. See the section titled "Approaches to the history of chemistry from a computational linguistic perspective," in Restrepo "Computational History of Chemistry – Part 2." Likewise, the State Key Laboratory of Organometallic Chemistry in China (discussed above) constitutes the case of an important laboratory for the recent practice of chemistry, in particular of organometallic chemistry (see n. 54).

citation networks, institutional affiliations, or the diffusion of terminology, we can identify hidden nodes of innovation, revealing how individual agency was embedded in broader structures, or we can even dismantle the very intellectual frameworks that have led history of science to write a progress narrative that tended to eliminate or miscontextualise information that indicated a different trajectory.[73] Conversely, macro-level patterns, such as the stabilisation of chemical discovery after 1860 or the rise of China in chemical discovery, cannot be understood without reference to the individuals, institutions, and practices that made them possible.[74] The macro is not above the micro; it is made of the micro. This is where the power of distant reading lies – not as a replacement for close reading, but as its necessary companion.[75] It allows historians to "see the forest" before the trees, identifying anomalies, structural shifts, or recurring motifs that then invite more focused inquiry. The real strength of computational history, as we see it, is not in choosing between scales, but in moving fluidly between them – for example by using the macro to generate questions, and the micro to answer them. Likewise using the micro to formulate hypotheses of the large-scale impact of particular events or individuals upon the historical unfolding of chemistry. In this way, the historian becomes not a passive observer, but a navigator on a path from patterns to meanings.

A further concern raised about the computational history of chemistry is the alleged fragility of digital data – epitomised by the disappearance of repositories, the obsolescence of file formats, and the breakdown of hypertextual links.[76] Yet, we maintain that these are not flaws of computation itself, but a mirror of the fragility inherent in all historical sources. A seventeenth-century manuscript may be damaged by fire; a nineteenth-century laboratory notebook may be lost in a move. The archaeological remains of early alchemical laboratories can get lost in time due to being repurposed.[77] The digital world is as fragile as the physical world. The difference is visibility: while the decay of a physical archive may go unnoticed until it is too late, the fragility of digital data is often detectable – and therefore preventable. We therefore insist that the solution lies not in abandoning digital tools, but in adopting sustainable practices: open access, version control with Git, rich metadata, and long-term preservation in trusted repositories like national libraries, Zenodo or OSF. Best practices such as data papers or Datasheets for Datasets help the creators and the

73 Examples of these studies include the computational history of chemistry results discussed by Restrepo in his Part 1 and Part 2 contributions to this *Ambix* issue. On what counts as knowledge, see: Lorraine Daston, "The History of Science and the History of Knowledge," *KNOW: A Journal on the Formation of Knowledge* 1, no. 1 (2017): 131–54, https://doi.org/10.1086/691678.

74 These results are also discussed in Restrepo's Part 1 and 2 in this Ambix issue.

75 On distant reading, see Franco Moretti, *Distant Reading* (London: Verso Books, 2013). On close reading, see Karine Chemla, ed., *History of Science, History of Text* (Dordrecht: Springer Netherlands, 2005), https://doi.org/10.1007/1-4020-2321-9.

76 This criticism is thoroughly discussed in Cohen and Rosenzweig, "Digital History." Further details on the fragility of digital data are found in Jeff Rothenberg, "Ensuring the Longevity of Digital Documents," *Scientific American* 272 (1995): 42-47; expanded in Jeff Rothenberg, *Ensuring the Longevity of Digital Information* (Washington, DC: Council on Library and Information Resources, 1999).

77 See Sarah Lang, ed., *Alchemistische Labore. Praktiken, Texte und materielle Hinterlassenschaften / Alchemical Laboratories. Practices, Texts, Material Relics* (Graz: Graz University Library Publishing, 2023).

users of datasets to audit and truly understand their contents.[78] These are not optional add-ons; they are the new non-negotiables for good scholarly practice. They ensure that data are both stored and that their provenance, curation, and limitations are documented. This transparency, if taken seriously in publication practices, is a profound advantage: it makes the research process visible, reproducible, and open to scrutiny. In this light, rather than a threat to trust, the fragility of digital data is an incentive for enhanced accountability. The digital archive is not a static monument; it is a living, evolving record of our engagement with the past.

Many of the challenges attributed to computational history are not unique, they are shared across all forms of historical inquiry. The problems of decontextualisation, the negotiation between different scales of historical inquiry ranging from microhistorical detail to macrohistorical patterns, and the fragility of digital data are not new. However, computational methods make these challenges visible, traceable, and, crucially, addressable. In this light, computational history is not so much a departure from the traditional tasks of the historian, but rather a way of tackling the same challenges with different methods. In our view, it demands a systematic engagement with the epistemology of data, the politics of representation, and the ethics of knowledge production. Far from undermining the historian's role, computational history cannot exist without it. The unique challenges that come with computationally-supported historical inquiry are invitations to rethink the discipline's commitment to, and stance on, rigour, reflexivity, and accountability. In short, the challenges of computational history are a call to reclaim and reinvigorate the historian's responsibility which can be easy to sideline when working on smaller scales and using tools approved by decades of historiographical practice.

The most glaring omissions of past histories reflect old biases that survive to the present: the diminution or outright exclusion of women, non-Europeans, and any person or group that is considered by the ruling class to deviate from norms policed by those in power. Indeed, such histories have hidden the existence of exceptional people who rose up despite the active oppression of "inferior races" and the "weaker sex."[79] Computational techniques provide historians with ways of

[78] On dataset documentation practices and auditing datasets, see Timnit Gebru, et al., "Datasheets for Datasets," *Communications of the ACM* 64 (2021): 86–92; Lang, "(Doing) Computational History"; Sarah Lang, "Documenting Datasets as a Tool for Change," in *Digital Humanities 2025: Book of Abstracts*, ed. Gimena del Rio Riande, et al. (Lisbon: ADHO, 2025), 885–87, https://zenodo.org/records/18393402 (accessed 3 May 2026). Sarah Lang and Elena Suárez Cronauer, "Dataset Audits for Mitigating Data Gaps," *Computational Humanities Research* (forthcoming 2026).

[79] See, for example, the classic Margaret W. Rossiter, *Women Scientists in America: Before Affirmative Action, 1940–1972* (Baltimore, MD: Johns Hopkins University Press: 1995). More generally see also James Poskett, *Horizons: The Global Origins of Modern Science* (London: Penguin, 2023); Hannah Wills, et al., eds., *Women in the History of Science: A Sourcebook* (London: UCL Press, 2023); Nina Rattner Gelbart, "Adjusting the Lens: Locating Early Modern Women of Science," *Early Modern Women* 11, no. 1 (2016): 116–27; Pamela H. Smith, "Science on the Move: Recent Trends in the History of Early Modern Science," *Renaissance Quarterly* 62, no. 2 (2009): 345–75, https://doi.org/10.1086/599864; Elizabeth Yale, "Teaching Women's Work and Thought in Undergraduate History of Science Courses," *History Compass* 21 (2023): 1–13. On women in chemistry in particular, see e.g. Marelene F. Rayner-Canham and Geoffrey Rayner-Canham, *Women in Chemistry: Their Changing Roles from Alchemical Times to the Mid-twentieth Century* (Philadelphia, PA: Chemical Heritage Foundation, 1998); Jeannette E Brown, *African American Women Chemists in the Modern Era*. (New York, NY: Oxford University Press, 2018).

uncovering some of what was hidden, seeing patterns of exclusion, as well as departures from such patterns, as we will now discuss.

Outlook: epistemic considerations of responsibility in a reflexive computational history

In the aforementioned focus section of *Isis* from 2019, Abraham Gibson and Cindy Ermus noted about their own experiences with computational history research, that "every step in the process drips with value-laden judgment calls."[80] The question of values embedded in research was brought to the forefront of the history and philosophy of science by Thomas Kuhn. His famous book, *The Structure of Scientific Revolutions,* highlighted a variety of tacit social and conceptual factors that influence theory change in the context of the history of science. Indeed, many of his longer examples focus on the history of chemistry.[81]

Since 1963, Kuhn's work has stirred controversy and encouraged critical examination of scientific practices, regardless of how one views his concepts of "revolution" and "paradigm."[82] The task of illuminating the broader social and cultural values that shape the actual practice of science fell to several schools of thought that emerged in the decades following Kuhn's *Structure*. While this is not the venue for an exhaustive review of those developments, two strands of that web are relevant here. One of these is the "Stanford school" of integrated history and philosophy of science ("& HPS"), whose influences are evidenced in the collection of articles in the previously discussed *Isis* focus section.[83] The other relevant strand of the web is the feminist philosophy of science.[84] Sandra Harding and Lorraine Code are most relevant to our discussion of persistent biases inherent in contemporary scholarship. We now turn to this strand.

80 Gibson and Ermus, "The History of Science," 58–59.

81 Kuhn uses examples from the history of chemistry throughout *Structure*, e.g., see pages 53–58 for Kuhn's sustained analysis of the "discovery of oxygen." Regarding tacit knowledge (i.e., knowledge that is not guided by explicit rules, but which eventually leads to the development of rules) Kuhn first raises the issue in the context of the priority of paradigms over rules, laws and theories in science (on 44), and how this is relevant to priority claims e.g., the discovery of X-rays, oxygen, and other phenomena. His 1969 "Postscript" (added to the second edition of Structure, in 1970) devotes an entire section, "Tacit Knowledge and Intuition, to this issue (on 191–98), see Thomas Kuhn, *The Structure of Scientific Revolutions* (Chicago, IL: University of Chicago Press), 1970.

82 The Max Planck Institute for the History of Science hosted a conference on the occasion of the fiftieth anniversary of Kuhn's book to explore its limitations and continued relevance: Alexander S. Blum, et al., *Shifting Paradigms: Thomas S. Kuhn and the History of Science*. (Berlin: Edition Open Access, Max Planck Institute for the History of Science, 2016). Also see, William J. Devlin and Alisa Bokulich, eds. *Kuhn's Structure of Scientific Revolutions: 50 Years On* (New York, NY: Springer, 2015).

83 Indeed, the main organisers of the *Isis* Focus section, Manfred Laubichler and Jane Maienschein, apply the Stanford school's &HPS framework directly in their research, for example: Jane Maienschein. *Embryos under the Microscope: The Diverging Meanings of Life* (Cambridge, MA: Harvard University Press, 2014); Manfred Laubichler and Gerd B. Müller, eds. *Modeling Biology: Structures, Behavior, Evolution* (Cambridge, MA: MIT Press, 2007). For an overview of the Stanford school see Section 5.1 in Jordi Cat, "The Unity of Science," in *The Stanford Encyclopedia of Philosophy*,ed. Edward N. Zalta and Uri Nodelman (Stanford, CA: The Metaphysics Research Lab, 2023) https://plato.stanford.edu/entries/scientific-unity/ (accessed July 7 2026).

84 Elizabeth Anderson, "Feminist Epistemology and Philosophy of Science," in *The Stanford Encyclopedia of Philosophy,* ed. Edward N. Zalta & Uri Nodelman (Stanford, CA: The Metaphysics Research Lab, 2024), https://plato.stanford.edu/archives/fall2024/entries/feminism-epistemology/ (accessed July 7 2026).

Given the massive volume of non-European alchemical archives, and the impracticability of their manual analysis, we hold that the deployment of computational methods is an "epistemic responsibility," to borrow Lorraine Code's terminology.[85] From Code's perspective, if it is possible for one to know *well* then it is imperative that one do so, and therefore wilful ignorance about the real world origins and consequences of knowledge is morally and epistemologically unjustifiable. in a computational history of chemistry context this would involve access to a broader range of data, enabling computational analyses that can uncover ways of knowing cultures whose organisational logic eludes standard Eurocentric models and expectations. Code's concept of responsibility indicates that knowers are not neutral, faceless epistemic agents, but rather individuals embedded in the cultures of their place and time, who are accountable for the impact of their modes of knowledge production on social and cultural values.[86] AI, a powerful tool of surveillance and control rooted in colonial systems, could also help restore respect for cultures shaped by colonisation, if the humanities take up this responsibility.

Sandra Harding's *Diversity and Objectivity: Another Logic of Scientific Research* responds to this call.[87] Her concept of "strong objectivity" entails that mere diversity is insufficient for moral and equitable decision-making regarding policy in general, and science and technology policies in particular. In her view, it is not enough to have high profile stakeholders, government officials and experts come together to discuss pertinent issues; the voices of the underrepresented, the powerless must also be heard. For example, in the planning phases of development projects, including the use and development of AI, key questions have centred primarily on the sustainability of profit and power for its financial stakeholders and secondarily on consumers.[88] Harding proposes that in addition to raising issues at stake for powerful players such as funders and directors, questions about maximising benefits and reducing harms to the least powerful should also be addressed. Indeed, strong objectivity results from facilitating radical diversity through the inclusion of groups or individuals who have been historically excluded at every step in the process.

For example, the post-World War II "Green revolution" opened foreign markets to US chemical industries, promoting the overuse of chemical fertilisers and pesticides in the name of fighting hunger, but without consulting farmers. The subsequent environmental damage mobilised reform movements. Some seventy years later, epistemic responsibility coupled with computational methods in chemistry have proven crucial for transforming restorative environmental justice into an empirically rigorous practice.[89]

[85] Lorraine Code, *Epistemic Responsibility,* (New York, NY: Oxford University Press, 1987).

[86] Jathan Sadowski, "When Data is Capital"; Sarah Lang, "Critical Concerns for Using LLMs."

[87] Sandra Harding, *Diversity and Objectivity: Another Logic of Scientific Research* (Chicago, IL: University of Chicago Press, 2019).

[88] Janet, Vertesi, Alex Taylor, and Benjamin Shestakofsky, "Reckoning with the Political Economy of AI: Avoiding Decoys in Pursuit of Accountability," arXiv preprint arXiv:2604.16106 (2026): 1–20; Pieter Verdegem, "Dismantling AI Capitalism."

[89] Xindi C. Hu, et al., "The Utility of Machine Learning Models for Predicting Chemical Contaminants in Drinking Water: Promise, Challenges, and Opportunities," *Current Environmental Health Reports* 10, (2023): 45–60. https://doi.org/10.1007/s40572-022-00389-x; Justin G. Teeguarden, et al., "Completing the Link between

We believe that digital and computational techniques, and AI's Large Language Models (LLMs) can play a beneficial role in addressing epistemic responsibility with respect to non-European languages and cultures, in the history of alchemy and chemistry and also more broadly. Such efforts would fall under the general category of AI-assisted restorative epistemic justice, a burgeoning area of research publications with a nearly 50% growth rate since 2020.[90] Similarly, AI-driven text recognition technologies have made great strides since 2022.[91] With such considerable gains in the capacity to read and translate premodern non-European text, it is now reasonable to expect a new beginning for the study of non-European alchemies. Efforts to actualise such potential will, however, require sustained collaborative interdisciplinary work.

In line with such efforts, "The Alchemy of Global Partnerships" workshop, held in May of 2025, brought together experts in digital humanities, library science, computer science, philosophy of science, and the histories of alchemy and chemistry to explore the transformative potential and persistent challenges of applying computational methods across cultural traditions. The workshop was convened by co-author Farzad Mahootian and hosted scholars from Abu Dhabi, Germany, the Czech Republic, China and the United States at NYU's Abu Dhabi campus.[92] Participants engaged in dynamic dialogues in person and online that revealed the need for an expansion of critical, historically grounded practice to shape the deployment of a variety of computational tools for the history of alchemy and chemistry.

The workshop was more than a fortunate convergence of the people, place and time; it was deliberately designed to address a disparity in the treatment of non-European sources in the history of alchemy, to cultivate collaborative interdisciplinary ideas in a zone of fruitful academic exchanges. The two studies of Ruland's *Lexicon* that appear in this special issue were initiated during the workshop; in fact, the use of AI and computational techniques to investigate Arabic terms in the *Lexicon* is the first of its kind in computational studies of the history of alchemy.

At present the standard material culture and textual sources of premodern science and technology are limited, and limiting. The case of non-European source materials for the history of alchemy presents an issue of "known unknowns," for these sources are, to an uncertain extent, uncatalogued.[93] Herein lies an opportunity for digital and computational techniques to assist in the process of transforming some unknowns into knowns more rapidly than previously conceived.

[89] *Continued*
Exposure Science and Toxicology for Improved Environmental Health Decision Making: The Aggregate Exposure Pathway Framework," *Environmental Science & Technology* 50(9): (2016): 4579–86.

[90] Soledad Zabala, et al., "Artificial Intelligence, Algorithmic Ethics, and Digital Inequality: A Bibliometric Mapping in the Digital Media Era" *Applied Sciences* 16, no. 6 (2026): 3056. https://doi.org/10.3390/app16063056.

[91] This area of research has also experienced a rapid growth curve, see Thea Sommerschield, "Machine Learning for Ancient Languages: A Survey, " *Computational Linguistics* 49, no. 3 (2023): 703.

[92] The workshop was generously supported by NYU-Abu Dhabi's Arts and Humanities Platform, NYU Office of Global Programs, and the Office of the Dean of Liberal Studies.

[93] Archives remain uncatalogued in private family collections, temple archives, and monasteries across North Africa, the Middle East, India and China.

However, the extent to which AI is useful for the purposes discussed here depends on the availability of well-curated and carefully annotated training data. Such resources are partly available for relatively modern contexts but are, generally speaking, far less developed for earlier historical periods and non-European languages. Furthermore, data modelling, preprocessing, and data cleaning, essential prerequisites for any subsequent "automated" analysis, tend to become more resource-intensive the older and less standardised the source material is.[94] The implicit standard for most digital processing remains contemporary English. To a lesser extent, many other Western and European languages are also comparatively well represented. As a result, the availability of data and models suited to particular research questions is generally greatest where multiple axes of power and visibility intersect, particularly in dominant Western languages and cultures and, to a lesser degree, in historical contexts regarded as especially significant within those traditions. At the same time, digital and computational methods also offer opportunities to address these imbalances. When applied critically, these methods can help accelerate the digitisation, processing, and cleaning of sources from historically under-resourced languages, cultures, and periods, thereby facilitating forms of research that would otherwise remain difficult to undertake at scale.

The computational history of alchemy and chemistry is still in its infancy. However its trajectory is clearly towards greater methodological transparency, interdisciplinary collaboration, and historical awareness of data. We hold that its future lies in critical engagement with the historian's data-algorithm pipeline, to question, to interpret, and to narrate. The historian is portrayed by some authors as one whose livelihood is endangered by technology and therefore must now "demand" a place at the table. For instance, Gibson and Ermus argue that "we should embrace the field's revolutionary potential, but we should not surrender what makes us unique. It is obvious that computers can analyse historical datasets, but they cannot *yet* interpret them. Therefore, we must *demand* a place for humans in the digital humanities."[95] Interpretation is a hermeneutical act that requires understanding, it is not something that instruments can do. In reality, it could be argued that computational history could not progress without historians. Based on the actual epistemic role of computational tools (rather than the marketing hype of the tech industry) it becomes clear that historians are not at risk of being displaced. There is no need to "demand" a place for them: the task is to develop clear

[94] It is important to note that the term "automated" may evoke an impression of simplicity and ease of use that is far removed from the reality of digital research workflows. Before any digital analysis can begin, substantial labour is typically required for data modelling, preprocessing, cleaning, and validation. For this reason, we place the term in quotation marks, as the apparent automation of the analytical stage can obscure the considerable amount of human labour that precedes it. This work can be computationally assisted but can never be fully automated. It would therefore be far removed from reality to assume that computational approaches simply consist of plugging existing data into an algorithm and obtaining a result almost instantly. In most cases, the required data do not already exist in a suitable form. Even when preliminary datasets or prior digital work are available, they can rarely be reused directly and typically require substantial modification, cleaning, restructuring, and validation before meaningful analysis becomes possible.

[95] Gibson and Ermus, "The History of Science" (emphases added).

judgement about where and how to exercise that place. Articulating that discernment is the job of historians. Gibson and Ermus point to algorithmic accountability and justice as two key areas to aggressively exercise humanity. We agree, and add epistemic responsibility and strong objectivity as broader areas of concern.

So, as we move forward, we must re-ask the fundamental question: What does it mean to know the past in the age of AI? This is not a technical question. As we have established earlier, it is a moral one. The history of chemistry has long been shaped by power structures: by those who had the resources to experiment, to publish, and to be remembered.[96] The digital archive, far from being a neutral mirror, reflects and reinforces these hierarchies. Computational methods offer a way to disrupt them, but we must be cognisant that those who shape advanced computational tools, including AI, are also those in power and as the well-known saying suggests, "the master's tools will never dismantle the master's house" (Audre Lorde). However, if complete "dismantling" is impossible, it is not unreasonable to redirect the tools to endeavours of knowing well and doing better for the marginalised and oppressed.[97]

AI is capable of both amplifying bias and revealing it. When trained on Western-centric corpora, it reproduces the same exclusions that have long defined the canon. But when trained on diverse, multilingual, and non-Western sources, it can detect patterns in alchemical traditions from China, India, Persia, and the Islamicate world that have been invisible to traditional scholarship. It can identify recurring motifs in untranslated manuscripts, flag underrepresented chemists and alchemists, and suggest new pathways for historical inquiry. This is not speculation; it is already happening.[98]

With power comes responsibility. If the future of the computational history of chemistry lies in co-creation of new methods and objects of knowledge, then collaboration must serve as its central methodological principle. As illustrated in the contributions of Hedesan, Pavlas, and Kaše; Lang, Pavlas, and Kaše; Lang, Mahootian and Lashen; and Restrepo, to this special issue, this means bringing together historians, chemists, computer scientists, physicists, linguists, philosophers, and language experts in sustained dialogue. The aforementioned workshop on the Alchemy of Global Partnerships was designed to facilitate just the interdisciplinary setting that is required to bring together the curators and annotators of data with the designers of algorithms, the framers of research questions. This is a form of

[96] Restrepo, "Computational History of Chemistry – Part 1."

[97] Pieter Verdegm, "Dismantling AI capitalism."

[98] Yannis Assael, et al., "Contextualizing Ancient Texts with Generative Neural Networks," *Nature* 645, no. 8079 (2025): 141–47; "Do Generative AI Models Output Harm While Representing Non-Western Cultures: Evidence from A Community-Centered Approach," https://arxiv.org/html/2407.14779v2 (accessed 17 February 2026); Moon-Kuen Mak and Tiejian Luo, "A Framework for Evaluating Cultural Bias and Historical Misconceptions in LLMs Outputs', *BenchCouncil Transactions on Benchmarks, Standards and Evaluations* 5, no. 3 (2025): NONE; Sarah Lang, "Fine-Tuning Machine Learning with Historical Data: An Alchemical Object Detection Dataset for Early Modern Scientific Illustrations," *Zeitschrift Für Digitale Geisteswissenschaften* 10 (February 2025); Lang et al., "Toward a Computational Historiography of Alchemy," *Proceedings of the Computational Humanities Research Conference 2023* (Paris: CHR, 2024): 29–48; "Bridging Traditions and Technology: AI in The Interpretation of Nusantara Religious Manuscripts," *Jurnal Lektur Keagamaan* 22 (2) (2024): 317–346, https://jlka.kemenag.go.id/lektur/article/view/1247 (accessed 17 February 2026).

research deliberately shaped by the interaction of diverse epistemic traditions. Indeed, an interdisciplinary computational approach to historical analyses constitutes a powerful way to reach beyond "digital colonialism" and allay the risk of replicating old power structures in new forms.[99]

Moreover, computational history is not just about analysing the past, it is about imagining new futures, by simulating counterfactuals: what if the periodic system had emerged in 1840? What if alchemy had been institutionalised in China rather than Europe? Scholars employing computational methods do not set out to rewrite history, but to test the robustness of narratives, explore the contingency of scientific development, and challenge the inevitability of Western scientific supremacy.

Every algorithm, every dataset, every model carries a history that must be interrogated, documented, and made visible. The history of the digital humanities offers cautionary lessons. Many early computational applications were exploratory, serving as experiments with data rather than as rigorous investigations of clearly defined research questions.[100] That is not to say that exploration is a bad thing. While the Computational Humanities Research community has since adopted a more hypothesis-driven approach (aligned with practices in computer science) this shift has, at times, led to a distancing from traditional humanities scholarship.[101] The methods and epistemologies differ so markedly that they can be difficult for traditionally trained humanists to follow, at first glance. We should energetically cultivate collaborative research settings, such as our Abu Dhabi workshop, that allow time for the natural development of a working communication flow – a pidgin language that enables understanding across disciplines without acquisition or relinquishment of expertise by either side. The workshop adapted Peter Galison's idea of interdisciplinary collaborative environments as "trading zones," an analogy with pre-industrial trading posts, places for producing transactions of value to all parties.[102] Such efforts to create sustained collaborative environments are necessary in order to prevent a wedge from forming between computational and traditional historians of

99 Toussaint Nothias, "An Intellectual History of Digital Colonialism," *Journal of Communication* 75, no. 5 (2025): 385–97. https://doi.org/10.1093/joc/jqaf003.

100 Sarah Lang, "Experiments in the Digital Laboratory: What the Computational Humanities Can Learn about Their Definition and Terminology from the History of Science," in *Fabrikation von Erkenntnis: Experimente in den Digital Humanities*, ed. Manuel Burghardt, et al. (Luxembourg: Melusina Press, 2022). https://doi.org/https://doi.org/10.26298/melusina.8f8w-y749-eitd.

101 https://computational-humanities-research.org/ (since 2020) and the journal Computational Humanities Research (CHR), Cambridge University Press, https://www.cambridge.org/core/journals/computational-humanities-research (accessed July 7 2026).

102 Peter Galison, a founder of the Stanford school of the philosophy of science, developed the notion of "trading zone" based on his field studies of interdisciplinary teams, following Bruno Latour, Steven Woolar, Karin Knorr Cetina, Sharon Traweek, and others. See, Peter Galison, *Image and Logic: A Material Culture of Microphysics* (Chicago, IL: University of Chicago Press, 1997); also see, Michael Gorman, ed., *Trading Zones and Interactional Expertise: Creating New Kinds of Collaboration* (Cambridge, MA: MIT Press, 2010). Most relevant to our discussion here is, Max Kemman, *Trading Zones of Digital History,* vol. 1 (Oldenburg: Walter de Gruyter GmbH & Co KG, 2021). For other case studies of embedded humanists in labs, see Michael Gorman, et al., "Integrating Ethicists and Social Scientists into Cutting Edge Research and Technological Development," in *Opening Up the Laboratory*, ed. N. Doorn, D. Schuurbiers, and M. Gorman (New York, NY: Springer, 2013): 157–73.

chemistry. The development of a niche practice of computational historians, detached from the broader community of the history of chemistry, would be most undesirable. The same holds for the history of alchemy, and the history of science generally.

This special issue is an invitation to historians, scientists, technologists, archivists, and to communities whose knowledge has long been excluded. It is a call to build histories of alchemy and chemistry that are more accurate and responsive to the achievements of cultures long occluded by orientalist prejudice. Let us not fear the machine, but transform it to enable more evidence-based investigations of cultures of discovery and invention on a global scale. And let us remember, the historian's most powerful tool is curiosity, the courage to ask: What if?

Lorraine Daston noted that, "[p]ursued from a cross-cultural and cross-historical perspective, the history of knowledge may well have consequences for our current classifications of knowledge and ways of knowing."[103] She went on to say that, "[i]n this wider context, the originary narrative of the history of science is not only false but also dangerous."[104] Multilingual and multicultural data sources, when included at scale, will ultimately impact the foundational narratives that shape the dynamic frontier between the history and historiography of science. The movement to decolonise education, to reevaluate and supplement existing narratives about the contributions of non-European cultures will gain the needed rigour and vigour to move the dial more rapidly than it has in the past half century. For which areas of the history of science, and by how much the dial will move, and how quickly, remains to be seen, but it surely depends on a more inclusive participation of scholars in the pursuit of these contested histories.

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[103] Daston, "The History of Science," 146.
[104] Daston, "The History of Science," 154.

Statement on the use of AI

During the preparation of this manuscript, the authors have used ChatGPT 4o and Qwen 3 Coder 30B A3B Instruct for proofreading and improving the English language. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

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